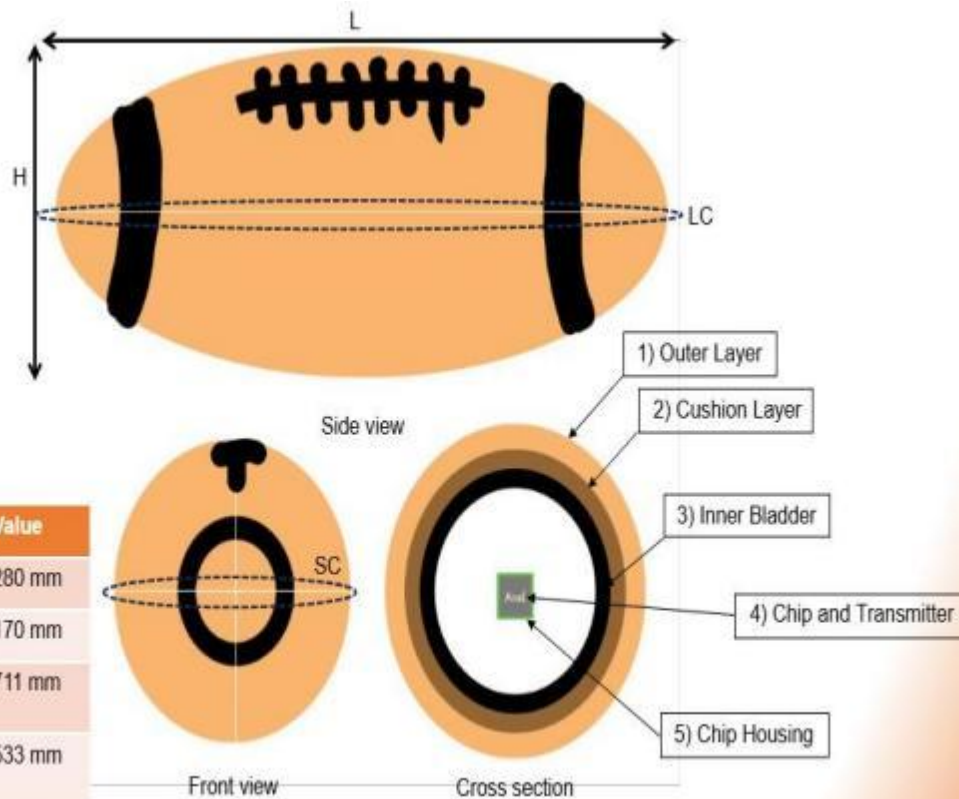


Final Report



A small, rechargeable power cell inside the core powers the sensor and transmitter. It supports all-game runtime and sealed charging (no open ports), so the ball stays water-resistant and safe during impact.

TEAM G-BMZ

TEAM MEMBER:

LUNCHENG ZHANG : OUT LAYER

MARTIN DRAKE: BLADDER

RYAN BERTELSEN: CHIP AND TRANSMITTER

Final Decision

Component	Objectives	Simplified Geometry	Free Dimension (Unconstrained)	Constraints	Designed (fixed) Variables
Out layer	1. minimize mass 2. minimize dollar cost (\$) 3. minimize sustainability cost (CO2 or EE or H2O) 4. maximize durability 5. enable efficient wireless charging	Spheroidal thin shell (non-conductive cover over the coil)	1. Outer Layer Thickness (especially above coil) 2. Material Choice 3. Coating choice	1. Shock absorption (hits, kicks, catches) 2. Failure under shear/torsion (occurs from throwing spiral) 3. Needs to keep air pressure 4. Keep NFL dimensions standards 5. Water resistant 6. Stress must be below material yield strength , and defotmation	1. Length L 2. Height H 3. Long Circumference 4. Short Circumference 5. Inner Pressure

Final Decision

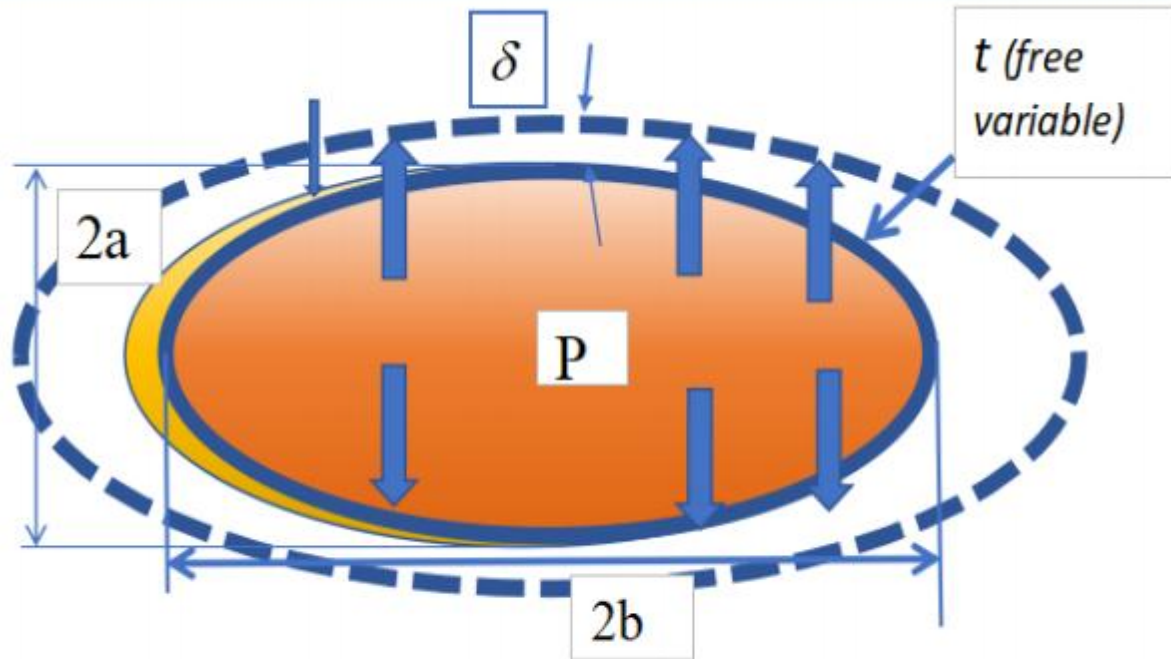


Figure 1: Simplified component geometry with loading conditions and design variables shown.

design variables		
p	82737.12	Pa
b	0.19	m
dmax	0.002	m
a	0.09	m

unit conversions	
GPa to Pa	1000000000
Mpa to Pa	1000000
m to mm	1000

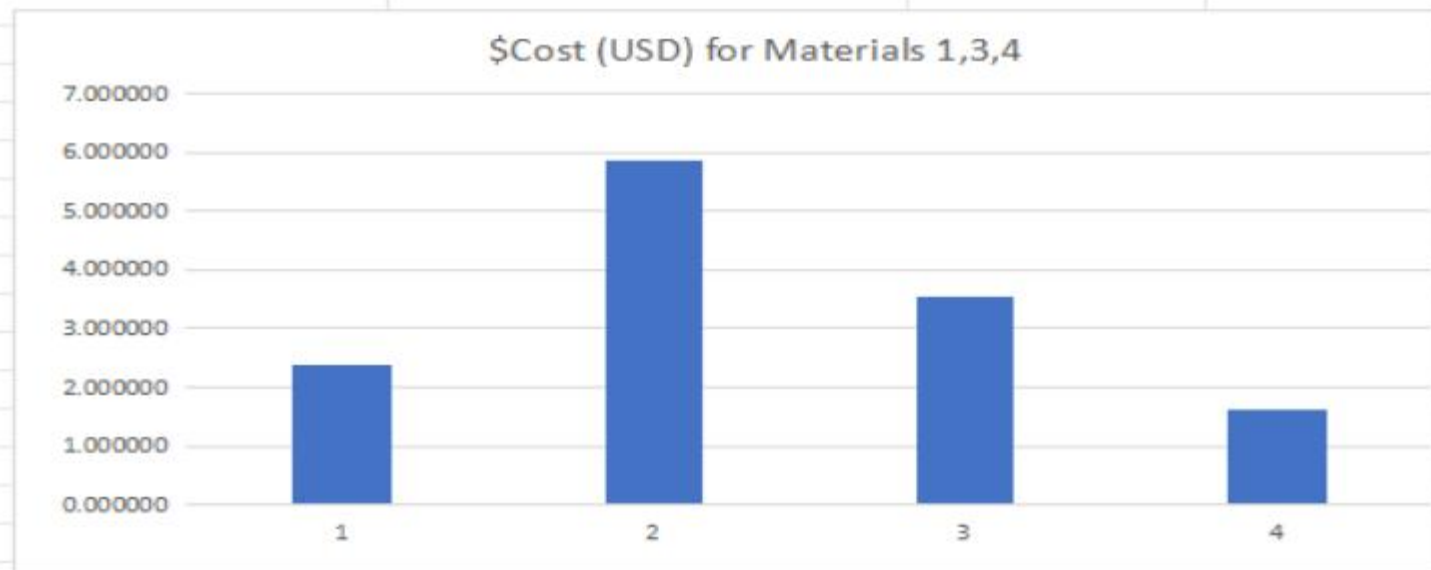
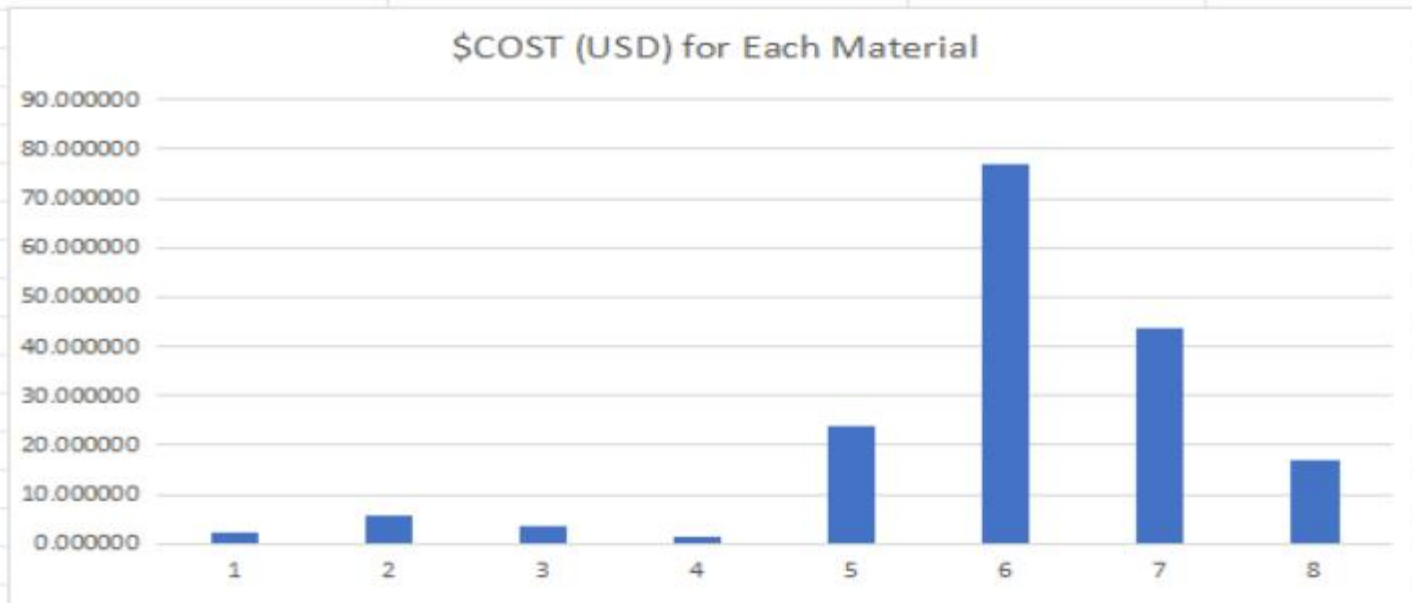
The main objectives for the outer layer focus on keeping the ball lightweight, affordable, durable, and functional with the internal wireless charging system.

A lower mass improves how the ball feels and performs during play. Minimizing dollar cost helps keep the manufacturing process practical for large-scale production. Reducing sustainability cost supports a more environmentally responsible material choice. Durability is also essential because the outer layer must survive repeated impacts and long-term outdoor use. Finally, the material must allow efficient wireless charging by remaining non-conductive and thin enough so the charging field can reach the internal coil.

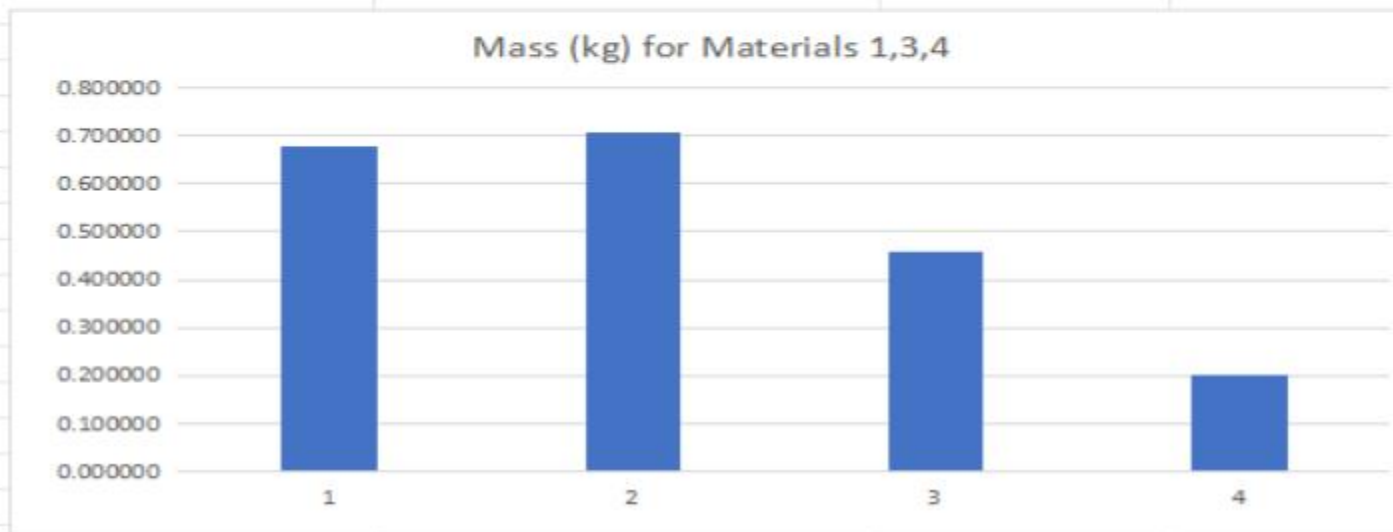
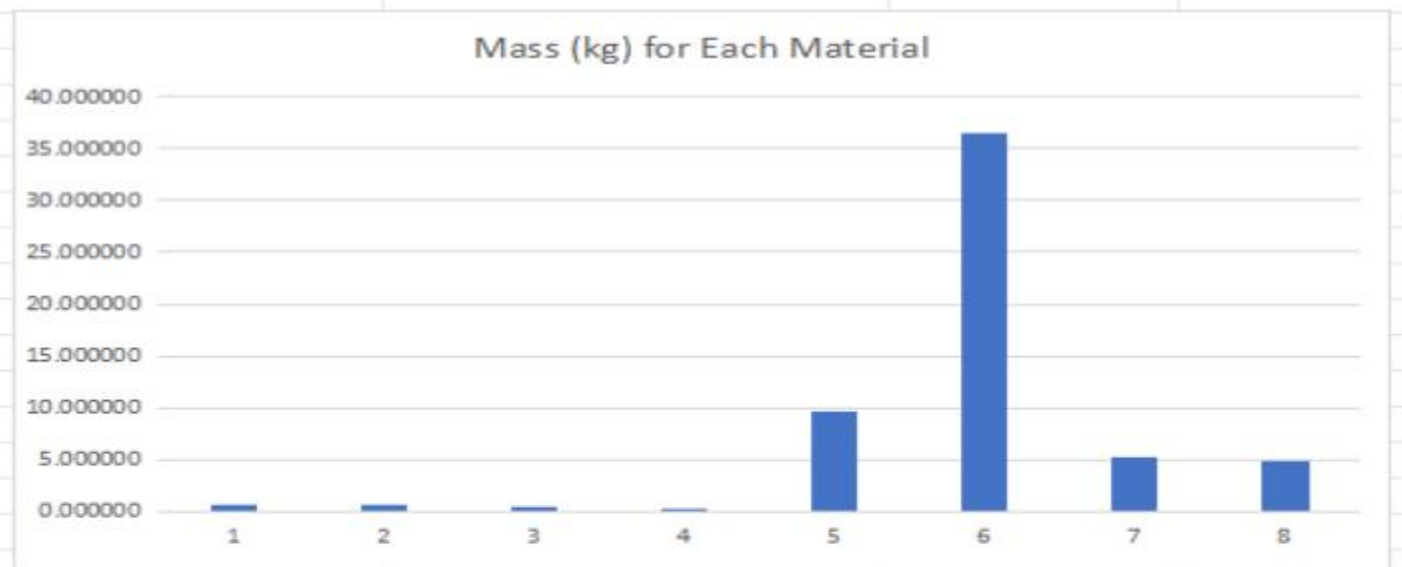
The outer layer must satisfy several strict performance constraints. It must provide enough shock absorption to protect both players and the electronics inside during hits, kicks, and catches. The material also must not fail under shear or torsion, which occur when the ball is thrown in a spiral. The outer layer needs to maintain air pressure by supporting the bladder and preventing deformation beyond the material's yield strength. NFL dimensions for length, height, and circumference must be met, so geometry cannot change.

Water resistance is required to protect the internal sensor system and to keep the ball safe in wet playing conditions. All stresses during use must remain below the yield limit of the chosen material.

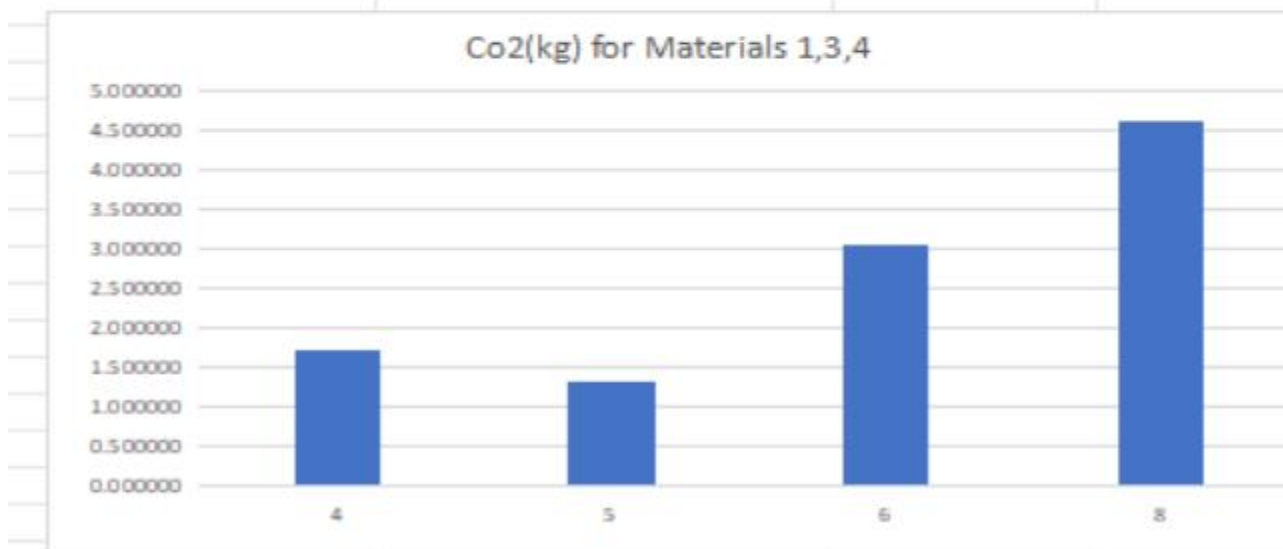
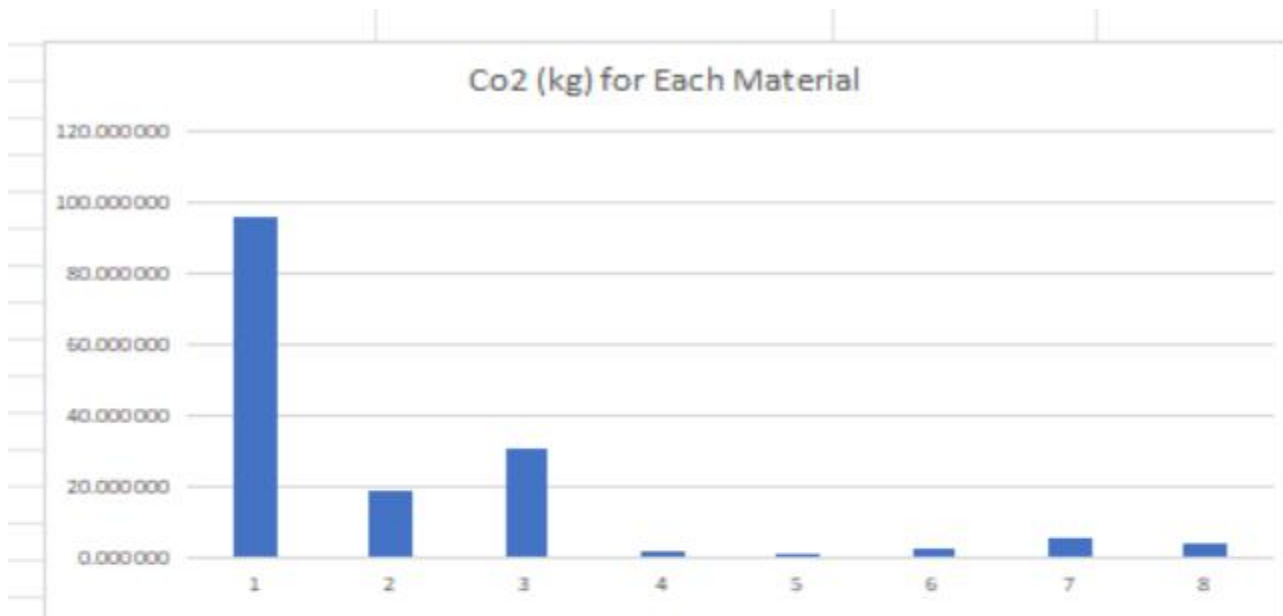
\$ cost Ranking



Mass Ranking

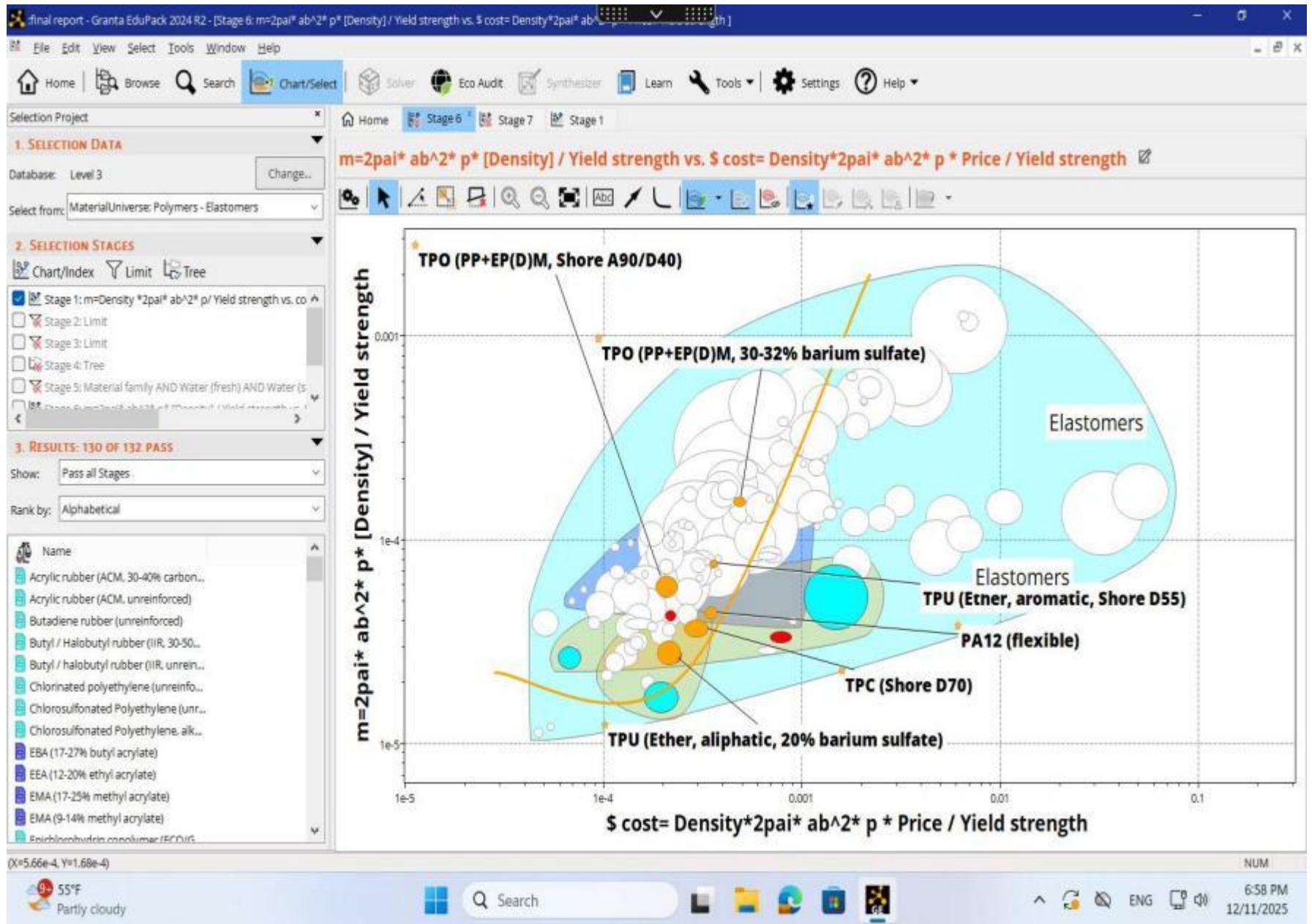


Co2 cost Ranking

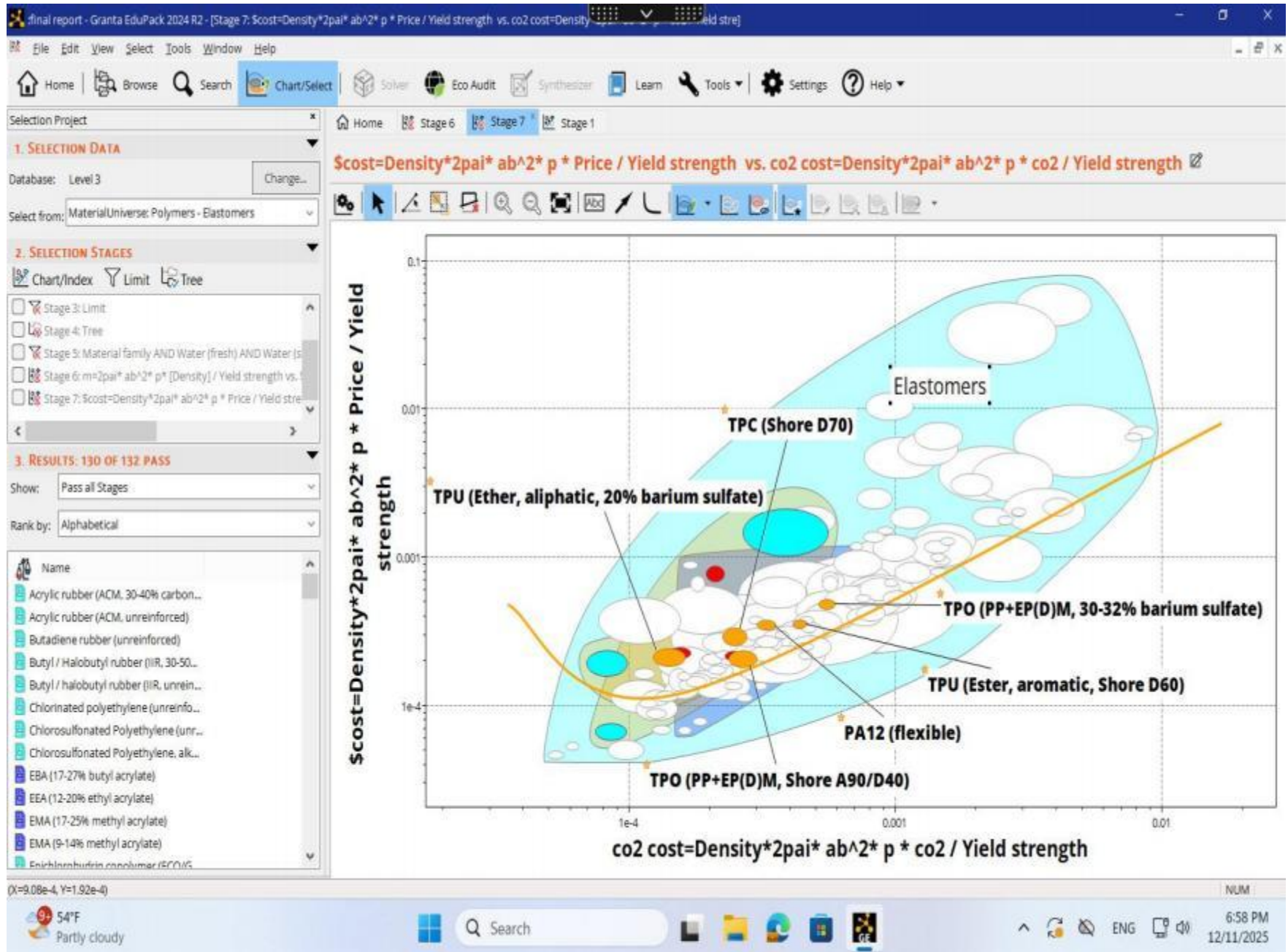


In the active constraint analysis I first used Granta to select polymer–elastomer candidates that satisfy the mechanical constraints for the smart football outer layer (deflection, yield strength, fatigue strength, and fracture toughness). For each candidate I then calculated the three objective indices – mass, dollar cost, and CO₂ cost – using the performance equations derived earlier, and plotted column charts to rank materials by the magnitude of each objective. Materials that produced extremely large objective values, such as very high mass, very high dollar cost, or very high CO₂ cost, were judged to require an unrealistic shell thickness or to be clearly non-competitive, so they were removed from the list. After this filtering, six materials remained for further study in the trade-off stage: TPO (PP+EPDM, Shore A90/D40), TPU (ether, aliphatic, 20% barium sulfate), TPC (Shore D70), TPU (ester, aromatic, Shore D60), PA12 (flexible), and TPO (PP+EP(D)M, 30–32% barium sulfate).

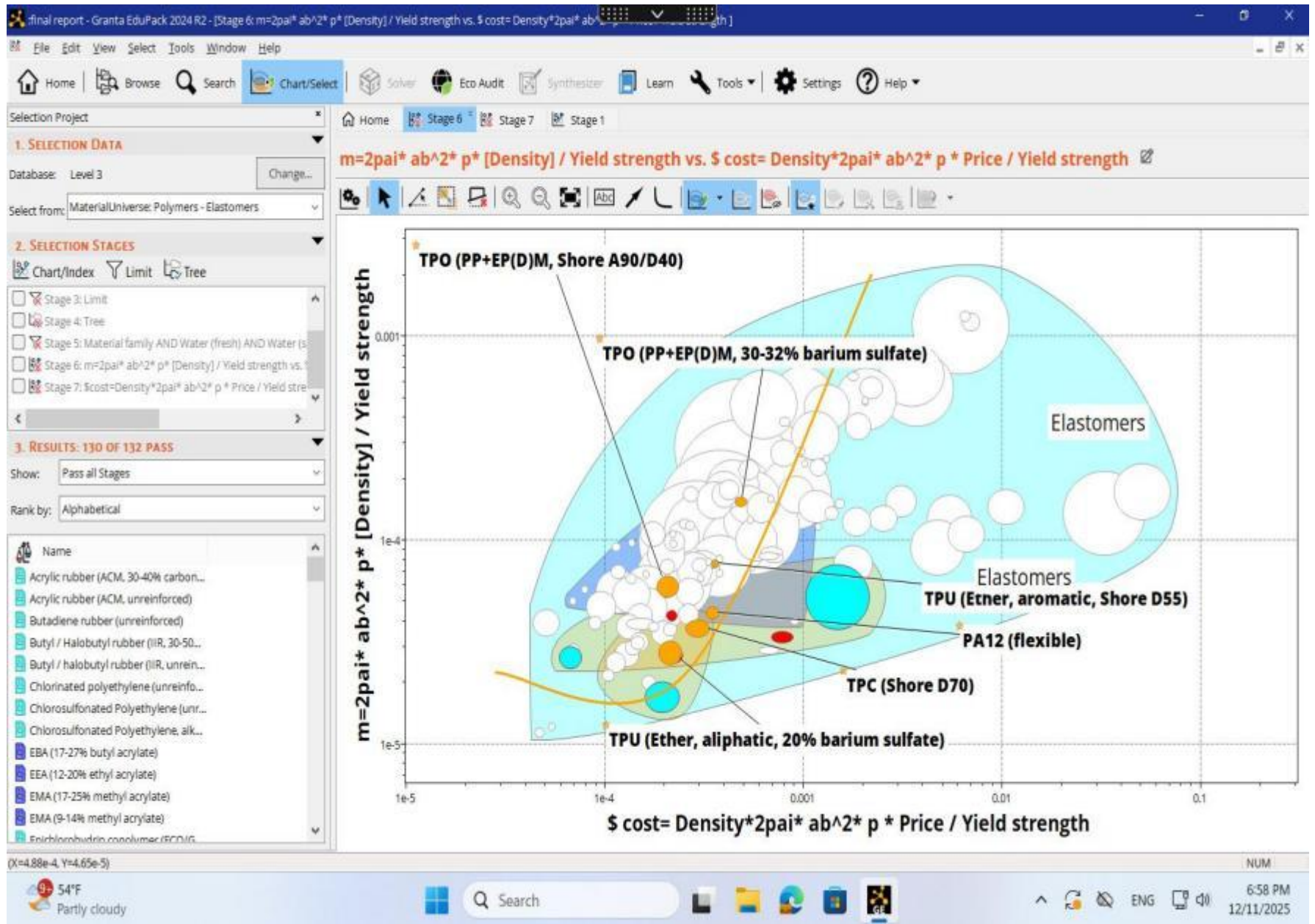
MASS/\$ COST



\$COST/CO2 COST



MASS/CO2 COST



The trade-off charts comparing mass, dollar cost, and CO2 cost show clear separation in performance among the six initial candidate materials. TPU- and TPC-based grades cluster near the lower trade-off boundary, indicating lower mass and cost per unit strength. In contrast, both TPO materials consistently appear higher on all axes, demonstrating that they require more mass and impose higher economic and environmental costs to achieve the same structural performance. Based on the plotted trade-off surfaces, the two TPO grades are identified as dominated materials because they show simultaneously higher mass, higher cost, and higher CO2 impact for comparable yield strength. PA12 is not dominated on the charts and is therefore retained for later evaluation.

After applying additional functional constraints—minimum Shore A hardness of 60, reliable water and oil resistance, impact durability, and compatibility with the wireless charging requirement—the feasible set narrows to three materials: TPU (Ester, aromatic, Shore D60), TPC (Shore D70), and TPU (Ether, aliphatic, 20% barium sulfate). These materials occupy favorable positions on the trade-off diagrams and simultaneously satisfy the mechanical and environmental criteria. PA12 is excluded at this stage because it does not meet the minimum hardness requirement, while both TPO materials were previously eliminated as dominated options.

Among the remaining candidates, TPU (Ester, aromatic, Shore D60) provides the most balanced combination of low mass index, low cost index, and sufficient durability for repeated impacts. TPC (Shore D70) offers strong mechanical strength but at a somewhat higher cost index, whereas TPU (Ether, aliphatic, 20% barium sulfate) remains viable but is penalized by increased density due to filler content. Considering the conflicting objectives and the added design constraints, TPU (Ester, aromatic, Shore D60) represents the most appropriate final selection for the outer layer.

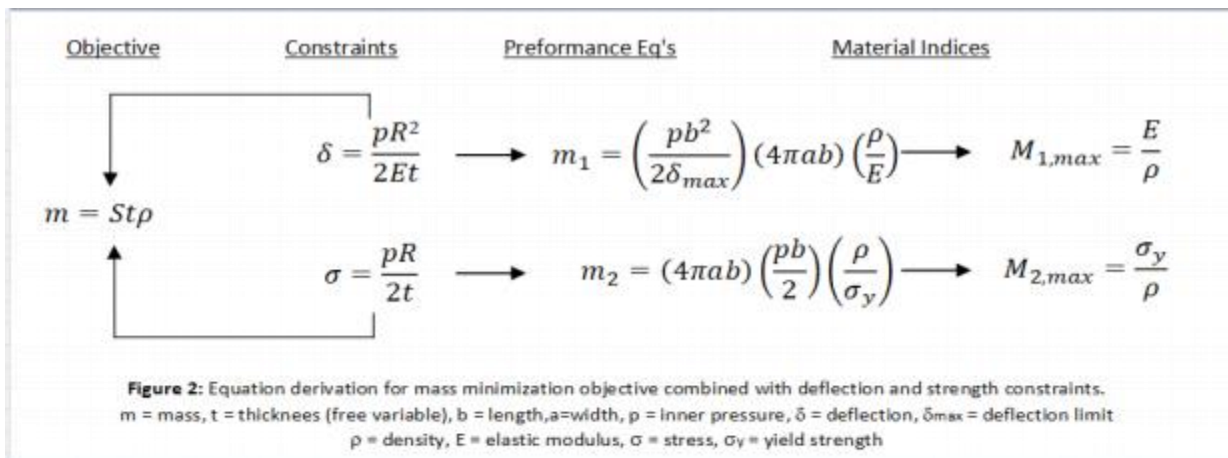
APENDNECES

Chart 1: \$cost and Mass material selection

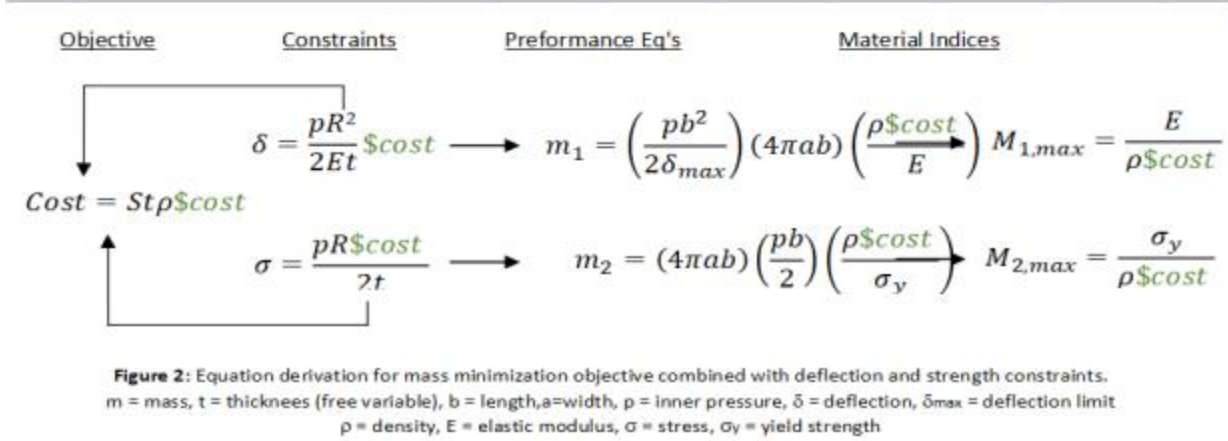
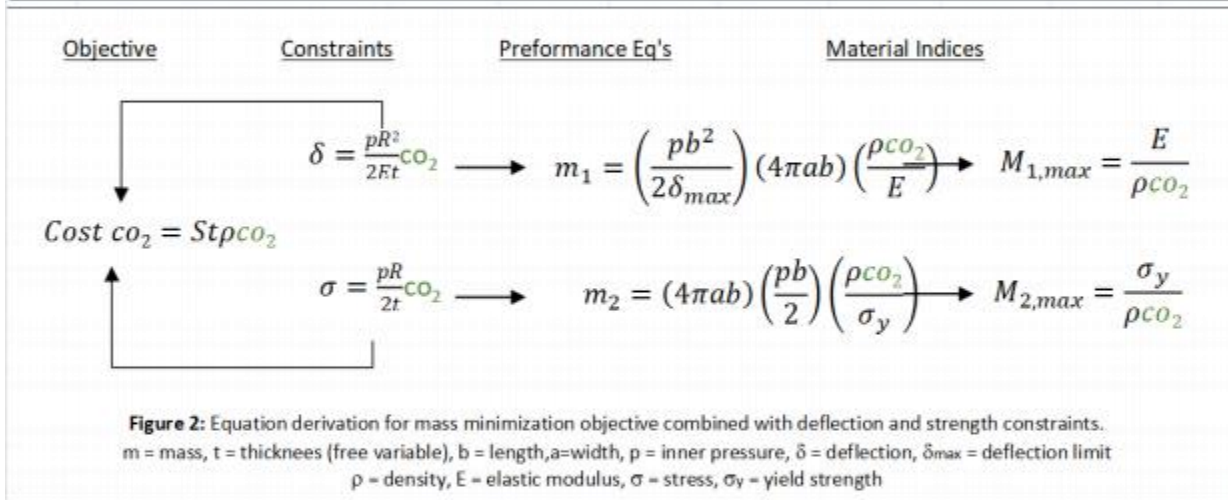
Objective related properties					Constraint related properties				
example material list	mass	cost	energy	co2 emitted	deflection	yield	fatigue	fracture	
	kg/m^3	\$/kg	MJ/kg	kg/kg	Gpa	Mpa	Mpa	Mpa*m^.5	
	density	cost/kg	energy	CO2/kg	E	σy	σf @ 10^7	K1c	
1 TPO(PP+EP(D)M, Shore A90/D40)	890		3.5	95	4.5	0.21	15	6.2	2.15
2 TPU(Ether,aliphatic, Shore D60)	1100		8.3	120	5.8	0.25	53	22.5	3.37
3 PA 12(flexible)	1030		7.7	142	7.5	0.36	24	16	2.65
4 TPC(Shore D70)	1250		8	130	6.7	1	34	14	9
5 Polyurethane rubber(unfilled)	1200		2.5	75	3.2	0.02	34	17	1
6 Styrene butadiene rubber(SBR,30% carbon black)	1140		2.1	110	1.9	0.005	20	8.5	0.32
7 TPU(Ether,aliphatic,Shore A80)	1080		8.3	120	5.6	0.033	38	15.5	1.5
8 POE/POP(Ethylene-based, Shore A80)	880		3.5	180	7	0.029	1.5	5	0.81

Chart 2: Co2 cost material selection

Objective related properties					Constraint related properties				
example material list	mass	cost	energy	co2 emitted	deflection	yield	fatigue	fracture	
	kg/m^3	\$/kg	MJ/kg	kg/kg	Gpa	Mpa	Mpa	Mpa*m^.5	
	density	cost/kg	energy	CO2/kg	E	σy	σf @ 10^7	K1c	
1 Styrene butadiene rubber(SBR,30% carbon black)	1140		2.1	110	2	0.0038	20	7	0.31
2 Natural rubber(15-42%) carbon black	1100		2.01	130	6.7	0.062	23	9.2	0.55
3 Polyurethane rubber(unfilled)	1200		2.5	75	3.2	0.02	34	17	1
4 TPU(Ester,aliphatic,20% barium sulfate)	1300		7.52	100	5	0.6	47	18	6
5 TPC(Shore D70)	1250		8	130	6.7	1	34	14	9
6 TPO(PP+EP(D)M, Shore A90/D40)	890		3.5	95	4.5	0.21	15	6.2	2.15
7 Ethylene Vinyl Acetate: 12% Vinyl Acetate	935		1.35	90	3.25	0.08	18.5	14	0.25
8 TPO(PP+EP(D)M, 30-32% barium sulfate)	1205		3.2	70	3.6	0.15	8	3	2



Figures: equations for object



performance - \$cost objective

deflection cost1 \$	failure cost2 \$	active max(1,2) \$ (USD)	rank low = best
2.3788777	0.350571	2.378878	2
5.8568504	0.290807	5.856850	4
3.5331234	0.557862	3.533123	3
1.6037378	0.496513	1.603738	1
24.0560667	0.148954	24.056067	6
76.7869648	0.202071	76.786965	8
43.5633498	0.398225	43.563350	7
17.0328012	3.466324	17.032801	5

performance - co2 cost objective

deflection m1 kg	failure m2 kg	active max(m1,m2) mass (kg)	rank low = best
0.6796793	0.100163	0.679679	3
0.7056446	0.035037	0.705645	4
0.4588472	0.072450	0.458847	2
0.2004672	0.062064	0.200467	1
9.6224267	0.059582	9.622427	7
36.5652214	0.096224	36.565221	8
5.2485964	0.047979	5.248596	6
4.8665146	0.990378	4.866515	5

free variable for mass and \$cost

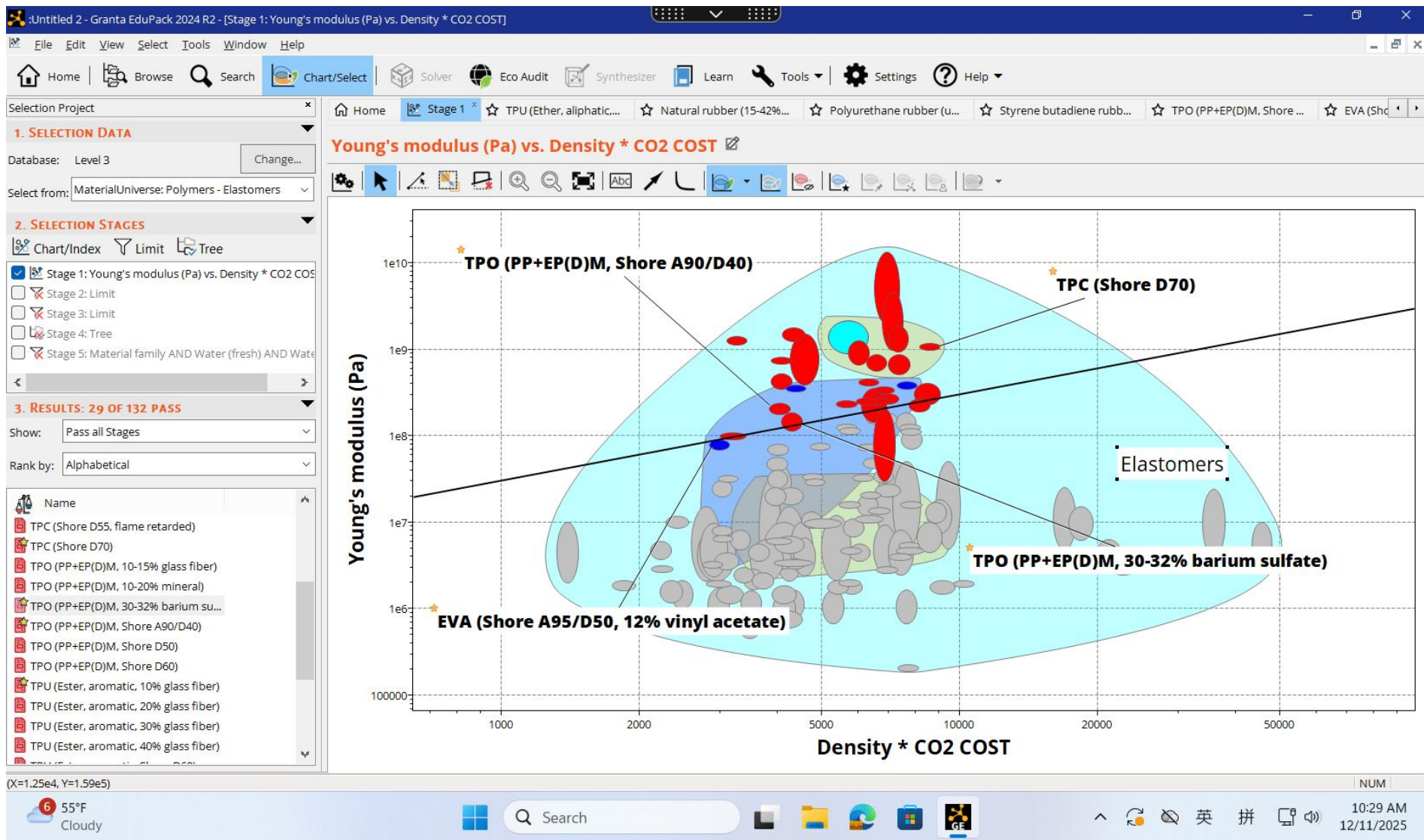
from deflection constraint	from failure constraint	active max(t1,t2) dia (m)
Thickness t1 m	Thickness t2 m	
0.00355573	0.000524002	0.0035557
0.00298681	0.000148302	0.0029868
0.00207417	0.000327501	0.0020742
0.00074670	0.000231177	0.0007467
0.03733513	0.000231177	0.0373351
0.14934050	0.000393001	0.1493405
0.02262735	0.000206843	0.0226273
0.02574836	0.005240018	0.0257484

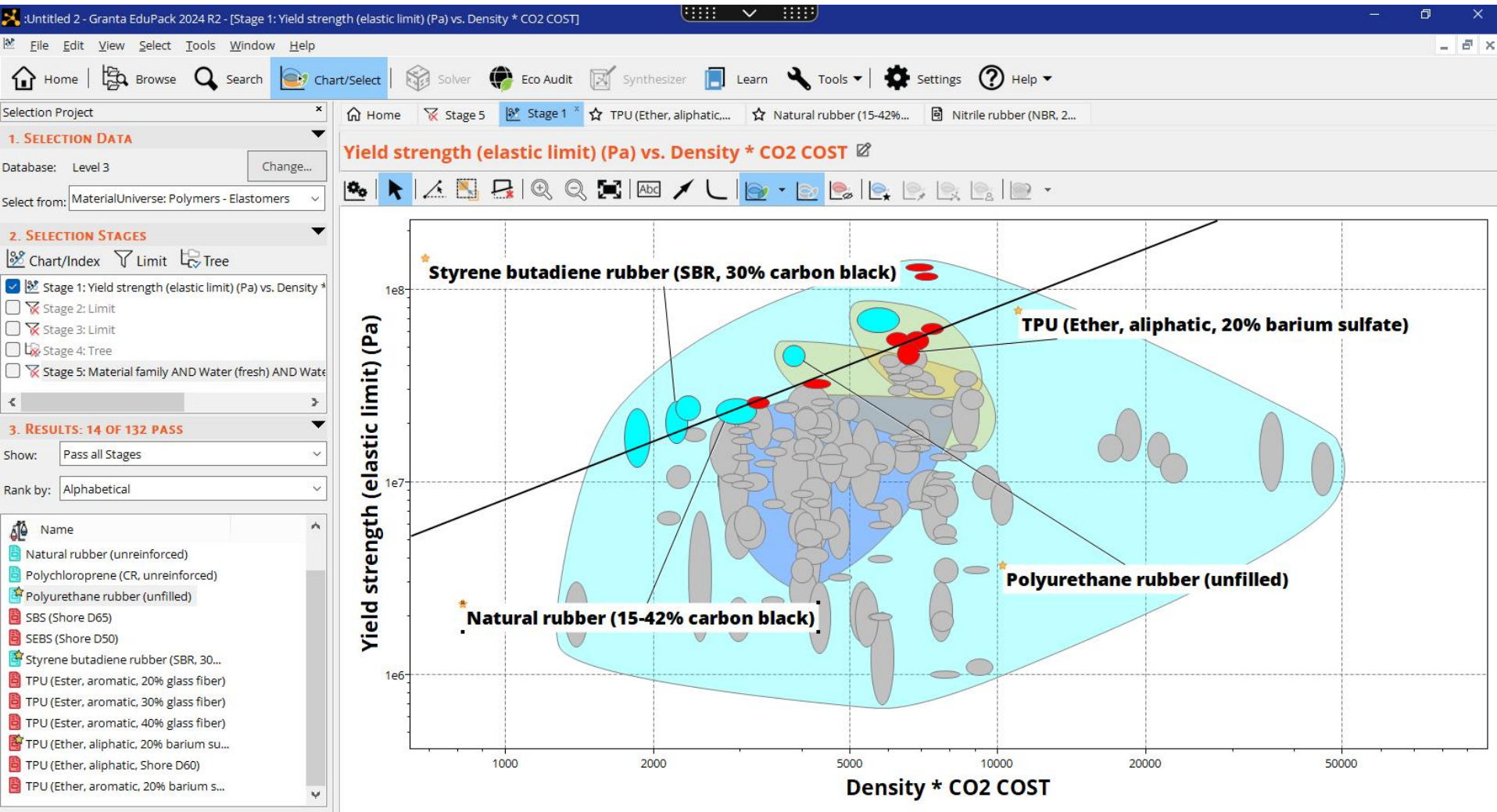
performance - \$ co2 objective

deflection	failure	active	rank
co2	co2	max(1,2)	low = best
kg	kg	mass (kg)	
96.2242667	0.192449	96.224267	8
19.0637862	0.540940	19.063786	6
30.7917653	0.190661	30.791765	7
1.7373826	0.233467	1.737383	2
1.3431304	0.062064	1.343130	1
3.0585570	0.100163	3.058557	3
6.0916977	0.085320	6.091698	5
4.6380097	0.254277	4.638010	4

free variable for co2 cost

from deflection constraint	from failure constraint	active
Thickness	Thickness	max(t1,t2)
t1	t2	dia (m)
m	m	
0.19650066	0.000393001	0.1965007
0.01204359	0.000341740	0.0120436
0.03733513	0.000231177	0.0373351
0.00124450	0.000167235	0.0012445
0.00074670	0.000231177	0.0007467
0.00355573	0.000524002	0.0035557
0.00933378	0.000424866	0.0093338
0.00497802	0.000982503	0.0049780





Ready

NUM

55°F
Cloudy



Search



10:16 AM
12/11/2025

Selection Project

1. SELECTION DATA

Database: Level 3 Change...

Select from: MaterialUniverse: Polymers - Elastomers

2. SELECTION STAGES

Chart/Index Limit Tree

☒ Stage 1: Young's modulus (Pa) vs. Density (kg/m³)

☐ Stage 2: Limit

☐ Stage 3: Limit

☐ Stage 4: Tree

☐ Stage 5: Material family AND Water (fresh) AND Water

3. RESULTS: 25 OF 132 PASS

Show: Pass all Stages

Rank by: Alphabetical

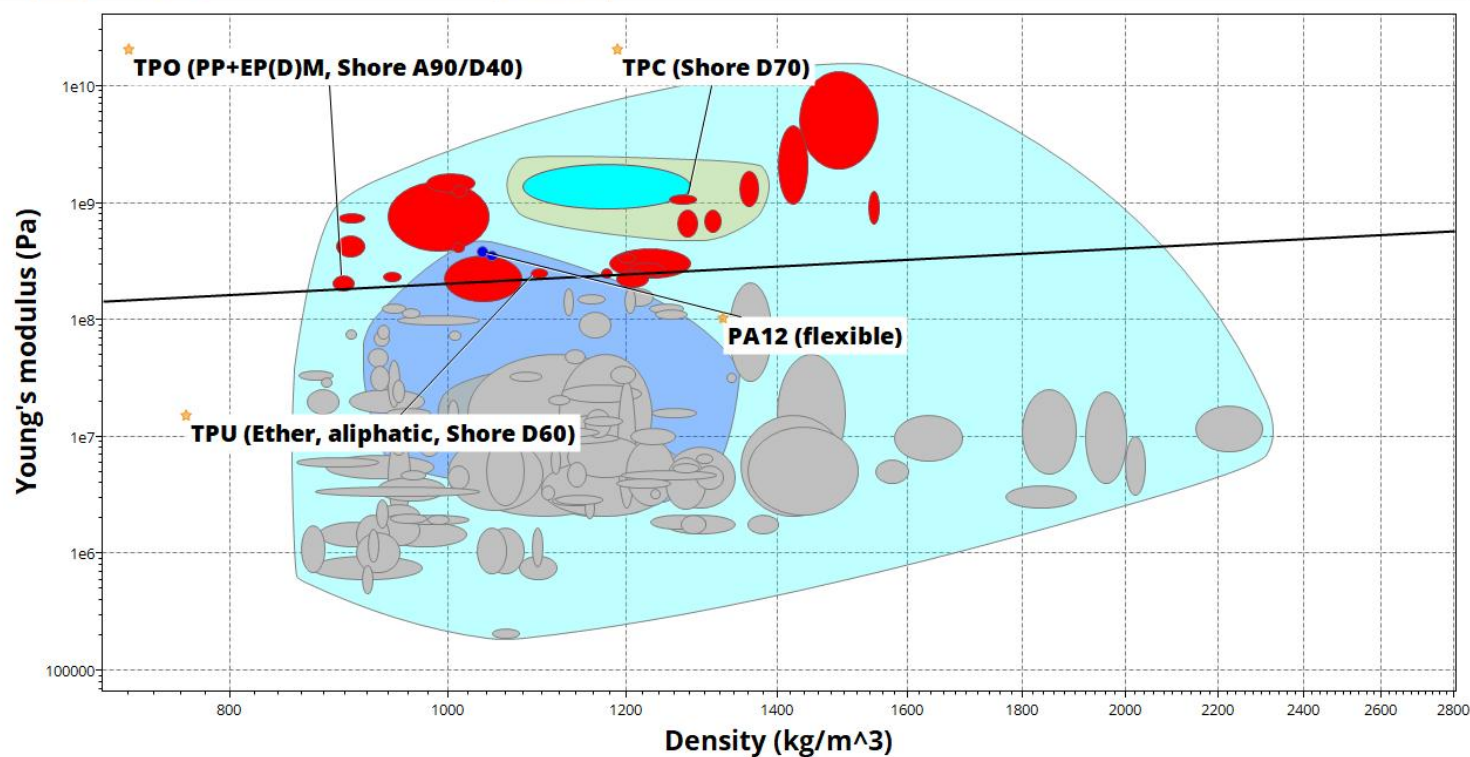
- Name
- Hard rubber (Ebonite)
- PA11 (flexible)
- PA12 (flexible)
- PEBA (Shore D55)
- PEBA (Shore D65)
- SBS (Shore D65)
- TPC (Shore D55)
- TPC (Shore D55, flame retarded)
- TPC (Shore D70)
- TPO (PP+EP(D)M, 10-15% glass fiber)
- TPO (PP+EP(D)M, 10-20% mineral)
- TPO (PP+EP(D)M, Shore A90/D40)
- TPO (PP+EP(D)M, Shore A90/D40)

(X=1.42e3, Y=6.61e4)

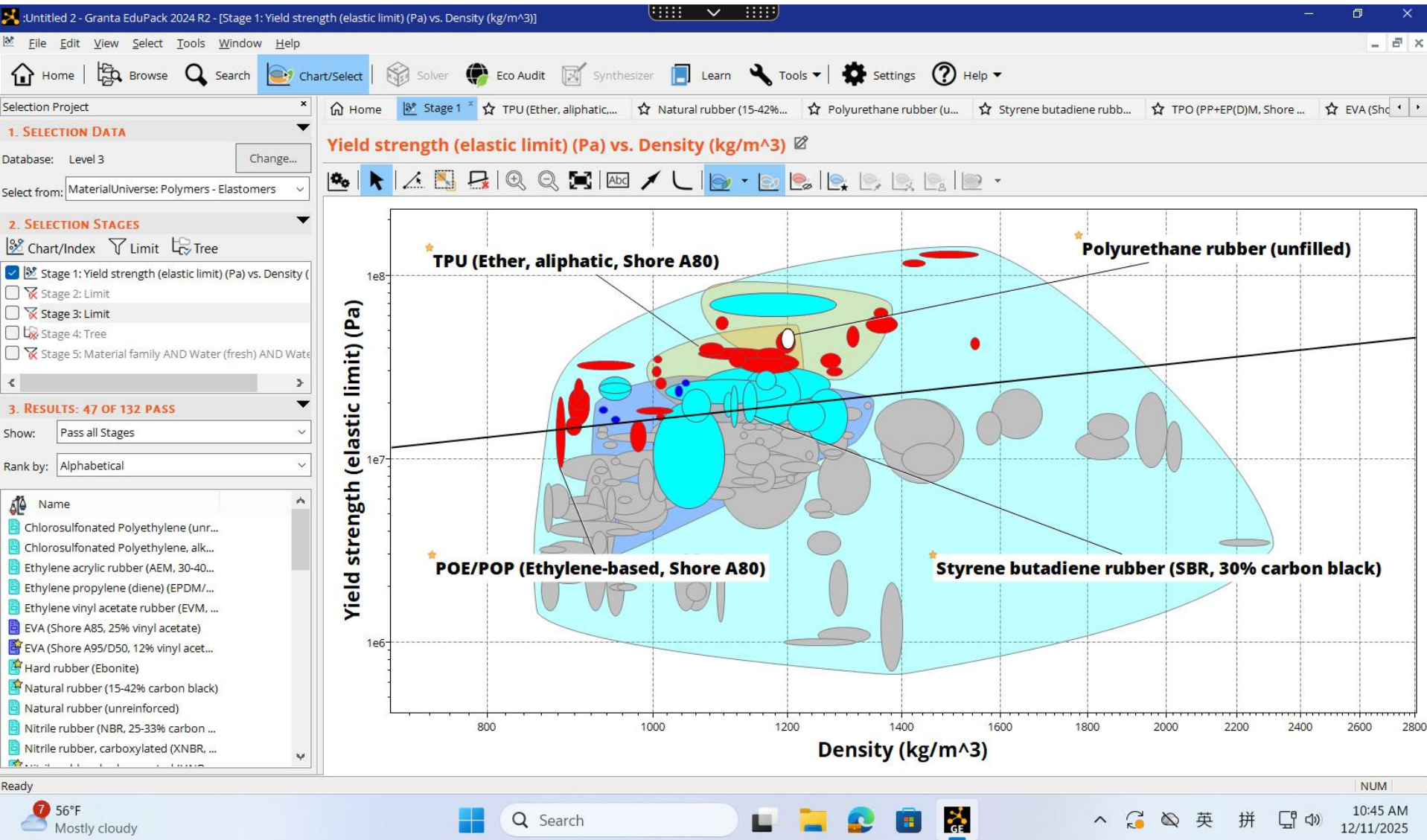
Home Stage 1 TPU (Ether, aliphatic, ... Natural rubber (15-42%... Polyurethane rubber (u... Styrene butadiene rubb... TPO (PP+EP(D)M, Shore ... EVA (Shd

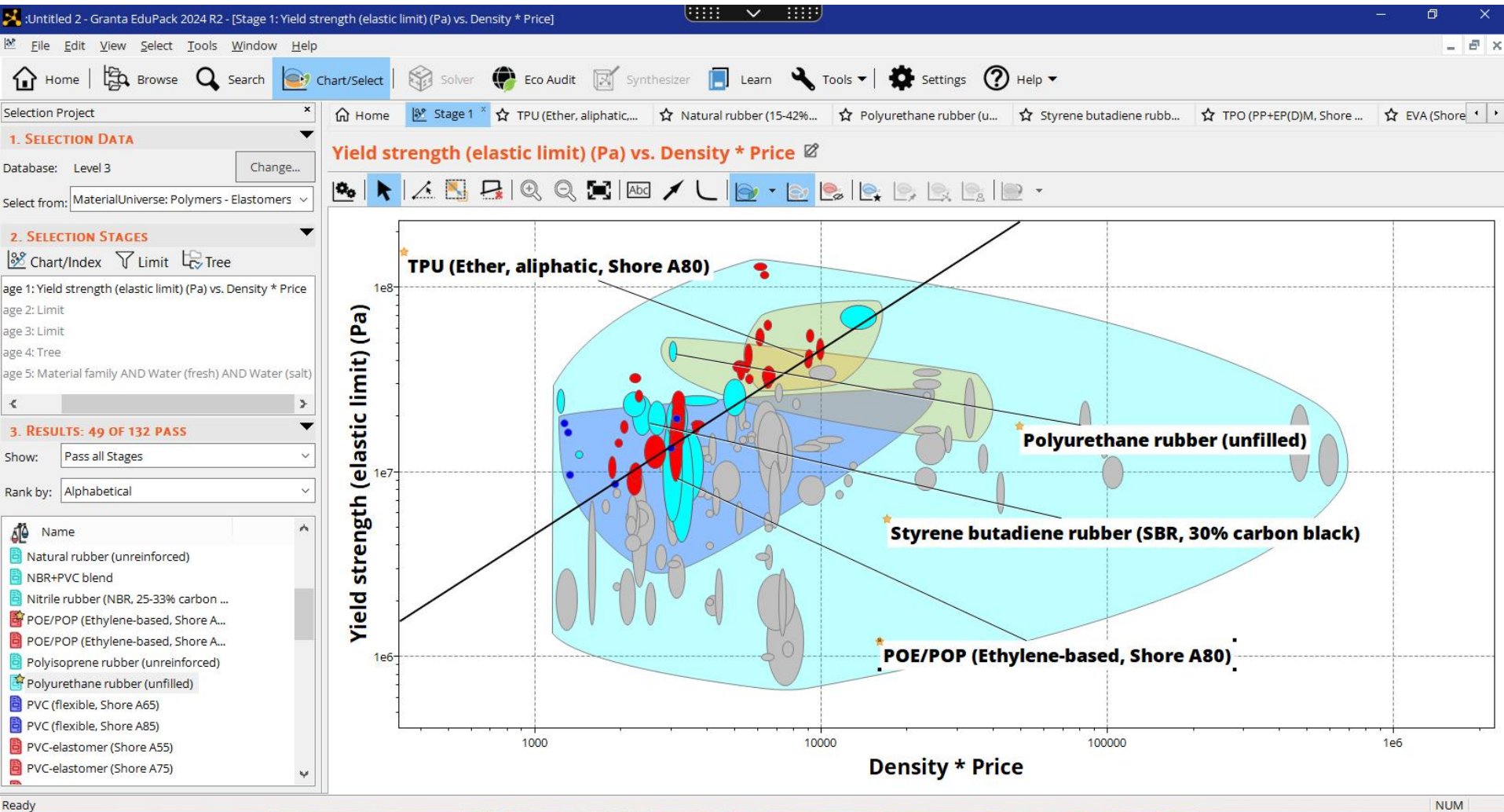
Young's modulus (Pa) vs. Density (kg/m³)

Chart/Select Solver Eco Audit Synthesizer Learn Tools Settings Help



(X=1.42e3, Y=6.61e4) NUM





Ready

NUM

Selection Project

1. SELECTION DATA

Database: Level 3 Change...

Select from: MaterialUniverse: Polymers - Elastomers

2. SELECTION STAGES

Chart/Index Limit Tree

age 1: Young's modulus (Pa) vs. Density * Price

age 2: Limit

age 3: Limit

age 4: Tree

age 5: Material family AND Water (fresh) AND Water (salt)

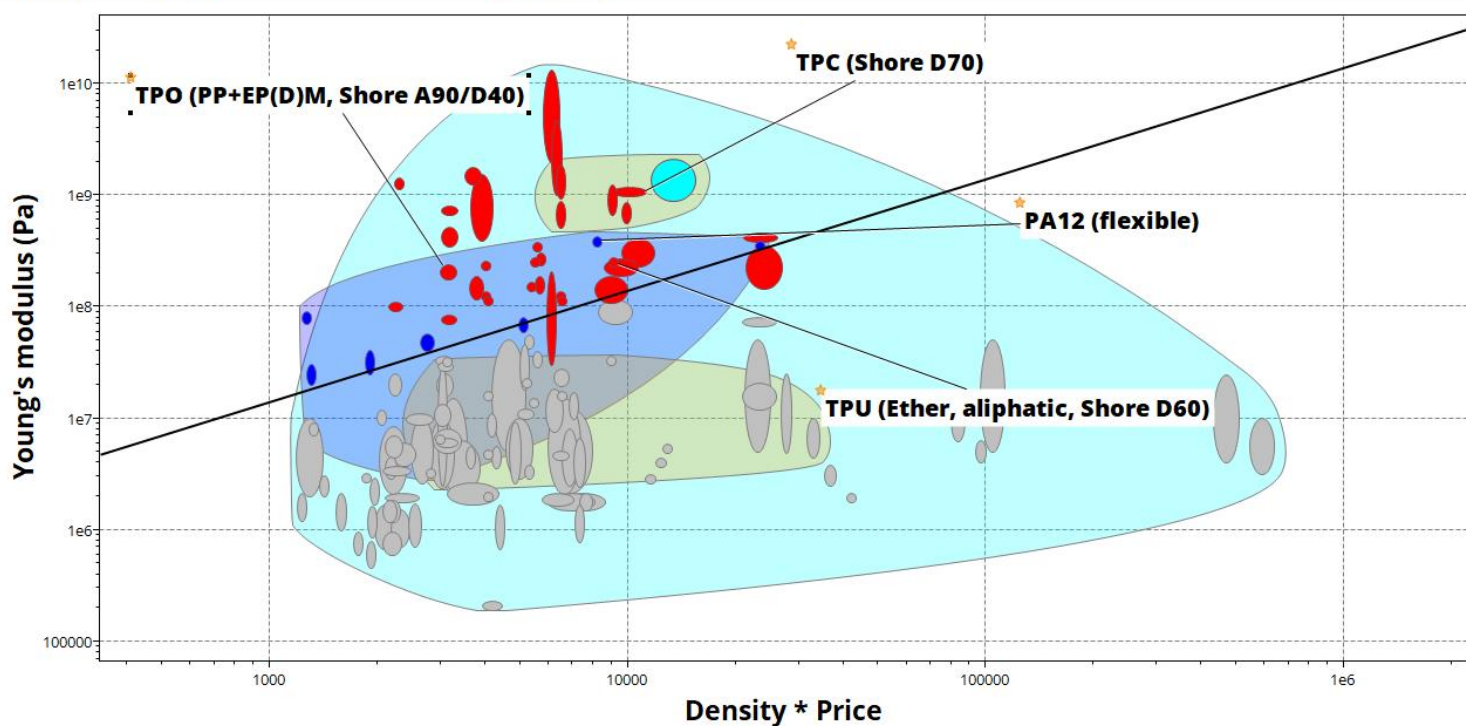
3. RESULTS: 41 OF 132 PASS

Show: Pass all Stages

Rank by: Alphabetical

- Name
- EBA (17-27% butyl acrylate)
- EEA (12-20% ethyl acrylate)
- EMA (9-14% methyl acrylate)
- EVA (Shore A85, 25% vinyl acetate)
- EVA (Shore A95/D50, 12% vinyl acet...
- Hard rubber (Ebonite)
- PA11 (flexible)
- PA12 (flexible)
- PEBA (Shore D55)
- PEBA (Shore D65)
- POE/POP (Ethylene-based, Shore A...

Young's modulus (Pa) vs. Density * Price



Ready

NUM